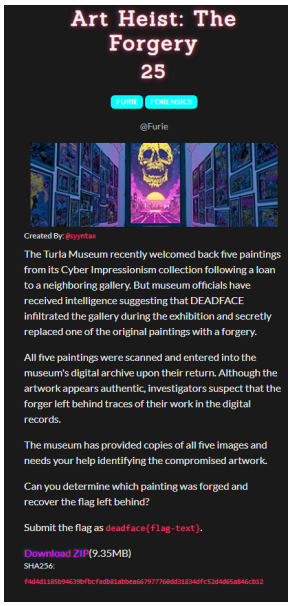
Solving “Art Heist: The Forgery” (25 pts - Forensics)

A step-by-step walkthrough of identifying a forged painting by inspecting PNG metadata and XMP fields.



Challenge page used during the solve.

1. The Challenge

Five scanned paintings were supplied. The goal was to determine which image had been tampered with and recover the flag left in the digital record.

2. Initial Triage

All five PNGs looked visually plausible, so the fastest path was to compare their embedded metadata rather than rely on visual differences.

- Inspect image dimensions and PNG metadata for all five files.
- Compare software, comments, XMP creator/title fields, timestamps, and identifiers.
- Look for one file whose metadata differs materially from the others.

3. The Anomaly

Painting_002.png stood out immediately. Unlike the other images, it reported **LAMIA Restoration Utility 0.0.13**, a restoration warning, and a suspicious XMP identifier.

```
Software: LAMIA Restoration Utility 0.0.13
Comment: checksum mismatch; palette recovered from cache
dc:identifier: deadface{L4MIA_SIGNS_HER_WORK}
```

4. Why This Confirms the Forgery

The other paintings used the normal archive metadata profile. Painting_002 contained a different tool name, a restoration-specific comment, a recent modification timestamp, and the flag directly in the XMP identifier. Those independent anomalies identify it as the compromised artwork.

5. The Flag

```
deadface{L4MIA_SIGNS_HER_WORK}
```

6. General Method

• For image-forensics challenges, inspect metadata before attempting heavier steganography.

• Compare every file in a set; outliers are often more important than any single field.

• XMP namespaces such as dc:identifier, dc:creator, xmp:ModifyDate, and software/comment fields are high-value targets.

Summary

Field	Result
Category	Forensics / image metadata
Compromised file	Painting_002.png
Key technique	PNG/XMP metadata comparison
Flag	deadface{L4MIA_SIGNS_HER_WORK}